

A Secure and Efficient Framework to Read Isolated Smart Grid Devices

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Abstract—With increasing deployments of smart grid systems, a large quantity of energy usage and grid status data have been collected by smart grid devices like smart meters. To secure these critical and sensitive data, it is crucial to prevent unauthorized readings from these devices. Many authentication protocols have been proposed to control access to smart grid devices that are a part of the smart grid data communication network; however, authentication protocols to control readings from the isolated smart grid devices are mostly ignored. In this paper, we propose a secure and efficient framework to enable secure data readings from the isolated smart grid devices based on a two-phase authentication protocol. The framework not only makes use of the smart reader as a bridge to connect the isolated smart grid device and the smart grid cloud, but also considers the physical constraints of all the devices in the system. Security analysis shows that our framework is efficient and secure under most typical attacks, meanwhile it satisfies the hardware constraints of smart grid devices. Comprehensive performance evaluation also validates the efficiency of the proposed framework.

Index Terms—Smart Grid, power system security, authentication, communication systems, data security.

I. INTRODUCTION

SMART grids have been proposed to improve the reliability, reduce the cost, and optimize the performance of the traditional power grid systems [8]. In recent years, many smart grid systems have been implemented and deployed by electricity service providers like OG&E [31]. In such systems, smart devices, like smart meters, play an important role in collecting data related to the power usage and the status of the smart grid; these data are generated by sensors such as plug-in electricity consumption monitor, load monitor and others. With a dedicated smart grid data communication system, these data are then transmitted to electricity service provider's data center [37].

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As described in several recent research efforts [18], [27], [28], [32], [36], by analyzing a large quantity of power usage data with fine granularity, utility client's personal privacy information, like daily activities, can be identified. It raises a significant concern on how to prevent unauthorized parties from reading the data collected and stored in the smart grid device. There are usually two types of smart grid devices, networked and isolated. The networked smart grid devices are a part of a smart grid data communication network, while the isolated smart grid devices exist in the area that is not covered by a smart grid communication network. Isolated devices are resulted from many causes including network outage, opt-out agreement between the electricity user and the utility company, and cost-effective considerations not to cover faraway devices. As a result, secure network protocol is usually used to read the data from networked smart grid devices, but the electricity service providers have to send a worker to the physical location of the smart grid device and read the data by using a handheld smart meter reader via a ZigBee-based communication protocol.

Secure frameworks are essential to prevent unauthorized readings from both networked and isolated smart grid devices. Recently several authentication protocols [2], [11], [29], [33] have been proposed to achieve this goal, but most of them are designed for networked smart grid devices only, and almost no effort has been made to limit the readings from isolated smart grid devices. In addition, because of the physical constraints of smart grid devices such as limited computation capability and memory space, heavy security schemes, like asymmetric key based algorithms, are not suitable to be applied in smart grid systems. This paper takes an initial step to develop a framework that can securely and efficiently read data from isolated smart grid devices.

The proposed framework makes use of the smart device reader (also referred as the reader) as the bridge to connect isolated smart grid devices with smart grid clouds. Firstly, the smart device reader gets authenticated with the electricity service provider's cloud through the cloud-reader authentication protocol. Then the cloud assists the smart device reader to securely generate a one-time symmetric key shared with the smart grid device, based on which the reader-device authentication protocol mutually authenticates the smart grid device and the reader. Security analysis of the proposed framework shows that the framework is free of most typical attacks to authentication protocols including eavesdropping, brute force attack, reply attack, device attack, internal attack, etc.

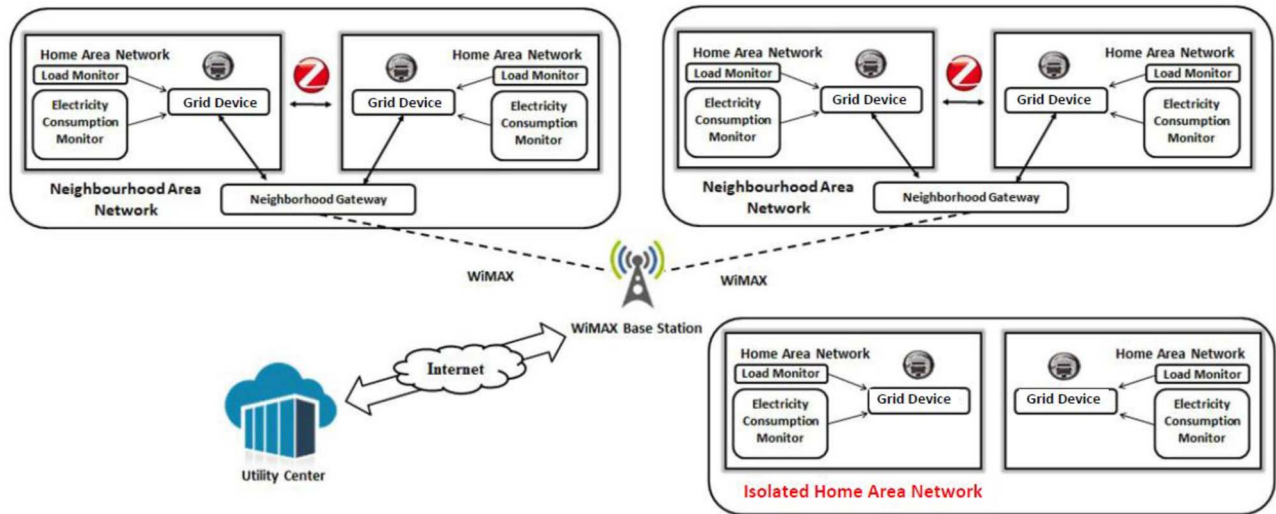


Fig. 1. Architecture of OG&E data communication network.

Performance analysis also shows that the symmetric key based protocol could have a much better performance than the asymmetric key based protocol, thus it is very suitable for resource constrained systems like smart grids.

The contribution of this paper is three-fold. First, we investigate the security issues of smart grids and design a novel framework to securely and efficiently read data from isolated smart grid devices. Second, we analyze the security properties of the proposed framework in the context of a set of typical attacks. Third, we evaluate the appropriacy of our design by a comprehensive performance evaluation.

The rest of the paper is organized as follows. An overview of a typical smart grid data communication architecture and the motivation of the paper are presented in Section II. The secure and efficient framework to read isolated smart grid device is proposed in Section III. We analyze the security properties of the proposed framework in Section IV, which is followed by the prototype and efficiency evaluation of the proposed framework in Section V. Section VI discusses a list of related work. Finally, conclusion and future work are depicted in Section VII.

II. SMART GRID ARCHITECTURE AND SECURITY CONCERNS

In recent years, smart grid has been proposed to improve the performance of the traditional power grid systems. It aims not only to integrate more green energy such as wind power and solar power, but also tries to improve the reliability and manage the load of the traditional power grid more efficiently. To achieve these goals, smart grid data communication networks that collect and monitor the energy usage and the status of the smart grid system have been built. Valuable applications have been developed based on the collected smart grid data. For example, energy usage information can help the smart grid system to distribute the load more wisely and design a fair but scaled pricing

model. Smart grid status monitoring helps identifying failures in the grid system and provides supports to fault-tolerant design. Automated metering infrastructure (AMI) [7], [16] is a typical approach to build the two-way communications between the energy user and utility. There are many ways to implement AMI as described in previous research including [6], [9], [15], [25], [26], and [38]. Another typical implementation of such a system from OG&E is shown in Figure 1. The system collects data from smart grid devices at residential homes or public buildings to the data center located at OG&E. The utility company, like OG&E, builds the AMI by themselves mostly because of several concerns such as needs of high bandwidth, more flexibility, better security and low cost.

Figure 1 shows the OG&E smart grid data communication network architecture, where several layers of communication are implemented. First, plug-in electricity consumption monitor, load monitor, smart thermometer and other smart grid devices (referred as device), like smart meters, consist of a home area network, which collects both power usage data and grid status data via a ZigBee based communication protocol. Currently, the data is collected at a frequency of one piece every 15 seconds. Furthermore, the smart meters and a neighborhood gateway form a neighborhood area network where the smart meters are self-organized into a smart meter ad hoc network and transmit the collected data to the neighborhood gateway using a multi-hop routing protocol. Moreover, the neighborhood gateways form the third layer network, a WiMAX network, which is connected to the forth layer network, a high speed public fiber network. The fiber network is responsible for transferring all the data to the data center located in OG&E. By analyzing the collected data, OG&E can design a time and location differentiated scaled but fair pricing model, can automatically bill the customer without sending an OG&E worker to check the meters at various locations, and can respond to any failures as early as possible by detecting the failures when they first appear. The reversed-way of

communication delivers the control message and notifications from the utility data center to the smart grid devices across the network.

Even though the ultimate goal of the smart grid communication system is to reach as many devices as possible and collect data from them, it is not possible to include every smart grid device into the two-way communication systems, even in the coming years, for the following reasons. First, based on the utility-scale smart meter deployments report [10] published in September 2014 from the Institute for Electric Innovation at Edison Foundation, as of July 2014, over 50 million smart meters have been deployed in the USA, but only covers over 43% of U.S. homes. By 2015 there are still 17 states with smart meter coverage less than 15% and 14 states with smart meter coverage between 15% and 50%. Please also note that not all smart meters are currently connected to a smart grid communication network. Moreover, based on the report, the increase of the number of new installed smart meters has slowed down in the last two years. Therefore, it will take many years to install and connect many more smart meters. We believe cost plays a significant role here, because based on the smart meter map provided in the report, less populated areas have a lower percentage of installation of smart meters. In other words, due to the cost of deploying the data communication network, it is not cost-effective to extend the data communication network to cover smart grid devices that are located sparsely and far away from the center of the network as depicted at the right-bottom corner of Figure 1.

In addition to the slow expansion of the smart grid network, there are also cases of natural disasters that will disconnect the smart devices from the smart grid network. For example, when a tornado strikes the Oklahoma City area, the above data communication network may be destroyed and it will take a while to be rebuilt. Third, for security concerns, some data collecting devices are not networked in the smart grid system as the scenario described in [35]. Last but not least, some utility users do not want to connect their smart devices to the smart grid network mostly because of security and privacy concerns. Therefore, there is opt-out policy provided by utility companies in most states having smart grid deployed. Based on news reports from Greentech Media [14] and others, there are about 1% users that are opt-out from the smart grid network. From all of the above, we conclude that we cannot eliminate the existence of the isolated smart grid devices, even though the utility companies keep replacing traditional power meters using smart grid devices like smart meters.

Regardless of whether the smart device is connected or isolated, the utility company always aims to read collected data from all smart grid devices efficiently and securely. In the isolated smart device case, OG&E sends workers to the physical location of smart devices and the workers collect the data by using a smart meter reader similar to smart phones and Tablets, via the ZigBee communication protocol. With smart meter installed, the worker does not need to walk close the meter and read it, but he can drive by the meter to read data. The readers carry the data back to OG&E data center.

In summary, The OG&E smart grid system collects data from two types of smart grid devices, the networked smart

grid devices and isolated smart grid devices, to improve the efficiency of the existing power grid system based on the data of detailed energy usage of appliances, the energy usage pattern of utility user, power quality related data, and meter data log and so on. With one piece of data collected every 15 seconds, considerable size of data is collected in a short time period such as one week or one month.

Unlike in previous years, the power grid device data only contained very simple information, like the total power consumption of the month. Nowadays, because of the fine granularity monitoring capability provided by using various sensors, much more data including the readings from plug-in electricity consumption monitor, load monitor and possibly other utility sensors are collected, transmitted, and stored at the smart grid devices. Based on several studies in last several years [18], [27], [28], [32], [36], researchers figure out that it is possible to identify a lot of utility client's personal private activities from the collected energy usage data. For example, from the energy usage data, researchers can answer a set of privacy-related questions such as when the people take a shower, when they cook, when they leave and go back home, whether they have kids, what type of disease they may have, and so on [28]. The privacy information implicated in the power usage data raises big concerns of privacy leakage if the smart meter data is compromised. Thus, secure protocols are crucial to prevent unauthorized reading to smart grid devices. In above described OG&E data communication network, it needs two sets of secure smart grid device reading protocols for connected and isolated smart grid devices respectively. In this paper, we focus on designing a framework to securely read smart grid devices for the latter.

III. DESIGN OF SECURE SMART GRID DEVICE READING FRAMEWORK

When an OG&E worker tries to read the data from smart grid device using a smart device reader, several types of attacks can be possible. First, an eavesdropper may listen on the communication channel between the smart grid device and the reader. Second, a fake device reader may be used to read the data from the smart grid device. Third, someone who is not working in OG&E may take an OG&E reader to read data. Fourth, an OG&E worker who is not assigned the task, but tries to use an OG&E reader to read the data from meter. Finally, the smart grid device may be modified to provide incorrect data.

To prevent all aforementioned attacks, we need to make sure that a legitimate worker is using a legitimate reader to read the data at the right time from a legitimate smart grid device located at the right location over a secure communication channel. Thus, we propose a secure and efficient framework to satisfy all above requirements. The rest of the section gives the details of the proposed framework. We first present the general idea of the proposed framework. Then, we give the design of two protocols that authenticate the reader to both the OG&E cloud and the smart grid device. Finally, the one-time symmetric key generation algorithm that establishes a pair of shared key between the reader and the smart grid device is discussed in detail.

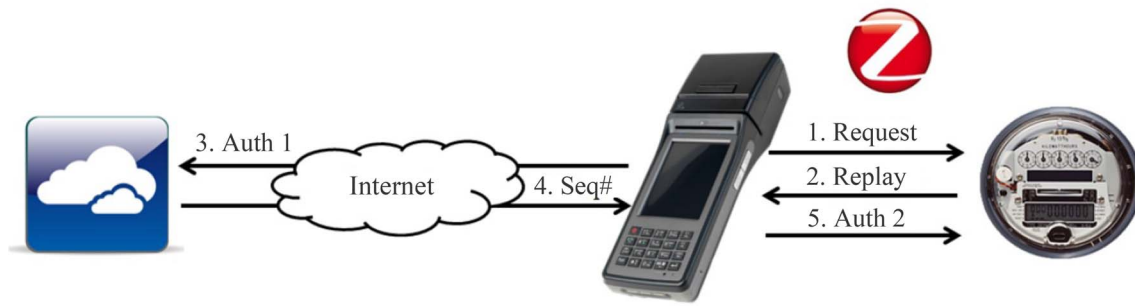


Fig. 2. Overview of the secure and efficient framework to read isolated smart grid devices.

A. Overview of Secure Smart Grid Device Reading

The overview of the secure smart grid device reading framework is presented in Figure 2, which involves three parties, the electricity service provider cloud (referred as cloud), the reader, and the smart grid device. Here the cloud is involved to help verify the legitimacy of the reader and assist the reader to get a new symmetric key shared with the smart grid device. The framework mainly consists of two phases, the reader-cloud authentication and the reader-device authentication. As depicted in the figure, there are five steps to be completed before the reader can read the smart grid device. First, the k_{th} round of data reading request is issued by the reader to the smart grid device. The smart grid device then responds by sending its ID in a message $\{M_{id}, TMSP\}_{SK_{k-1}}$, where M_{id} is the smart grid device ID and $TMSP$ is the timestamp of this message. The message is encrypted by the $(k-1)_{th}$ round key shared between the device and the reader or by predefined key if this is the first session. Next, the reader starts reader-cloud authentication by providing all information including smart grid device ID, worker information and other information such as location from GPS and reader ID. After the cloud verifies the legitimacy of the reader based on the information received, the cloud provides necessary information that helps to generate a new shared key between the reader and the smart grid device. Then it starts the second phase of the framework, the reader-device authentication. Successfully completing all above steps, the reader is then allowed to read the data from the smart grid device. We present the details of two authentication protocols in the coming section.

We adopt a two-phase framework involving three parties because of the following several considerations. First, the smart grid devices are usually not powerful enough to support heavy computations needed in some cryptographic algorithms like asymmetric key schemes. Thus, a lightweight authentication protocol is suitable to be applied here. Moreover, isolated smart grid devices may not have enough knowledge to check the legitimacy of the reader without the help of others. Finally, the two-phase framework enables distributing the storage of the information used in authentication such as the previous shared keys and the data used to generate the new key needed in security protocols so it can be more resistant to different attacks. Therefore, the OG&E cloud is a part of the secure smart grid device reading framework and assists the reader-device authentication.

B. The Reader-Cloud Authentication Protocol

Each time the reader tries to read the smart grid device, the cloud helps to verify that a legitimate reader with a legitimate worker is accomplishing an assigned task at the right time so that several aforementioned attacks can be blocked. An asymmetric key based protocol is used in reader-cloud authentication as depicted in Figure 3, because readers, like Tablets, are powerful enough to support asymmetric key based security schemes. Here, we assume that both the reader and the cloud have each other's public key. This can be easily achieved, because the reader is a part of the cloud before the OG&E worker takes it out to work.

The reader-cloud authentication starts after the reader gets the ID from the smart grid device. Figure 3 depicts the details of the authentication protocol. The reader first sends an authentication request message $\{\{R_{id}, T_R\}_{Key}, \{Key\}_{Pub(c)}, \{R_{id}, T_R\}_{Pri(R)}\}$, where R_{id} is the reader ID; T_R is the reader's request timestamp; Key is the one-time session key to encrypt the rest of the communications between the cloud and the reader in this session; $Pri(R)$ is the reader's private key; and $Pub(C)$ is the cloud's public key. The message contains the signature of the reader with reader's request timestamp, the encrypted reader ID and a timestamp using Key , along with the encryption of the Key using the cloud's public key. After receiving the message, the cloud decrypts the Key using the cloud's private key. With the Key , the cloud can decrypt $\{R_{id}, T_R\}_{Key}$ and get the reader's ID and timestamp. By comparing the decrypted reader ID and the timestamp, the cloud can verify the signature of the reader. After the reader identity verification, the cloud responds with message, $\{C, T_R\}_{key}$, where C is the cloud ID, T_R is the reader's request timestamp, and Key is the same encryption key the reader has used in the first message. Receiving the response message, the reader can verify the identity of the cloud, because only the right cloud can get the Key so as to get the T_R from the reader. Then, the reader sends task related information including the worker ID (U_{id}), the smart grid device ID (M_{id}), the reader's location (Loc), and the working time ($Time$), in a message $\{U_{id}, M_{id}, Loc, Time\}_{key}$, encrypted by the same session key. Next, the cloud checks the received task information with the worker schedule calendar stored in the cloud. For example, the cloud checks if the worker ID matches the reader ID, if the location of the meter is within the communication range of the smart grid device; and if the

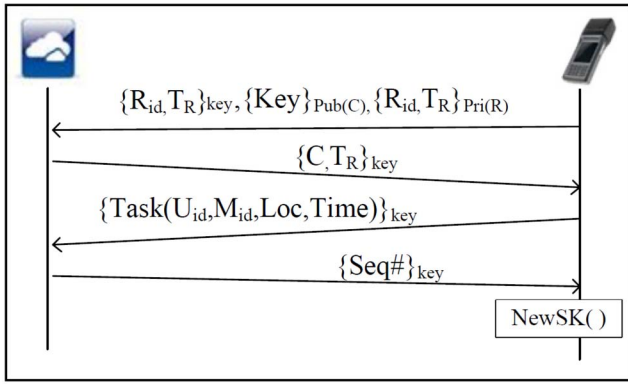


Fig. 3. The reader-cloud authentication protocol.

task is performing at the assigned time slot. After verifying that the reader is authorized to read the smart grid device that is located in the specified location at a certain time based on the cloud's database, the cloud authenticates the reader and sends the sequence number related to that specific M_{id} to the reader, in the message $\{Seq\# \}_{key}$, where $Seq\#$ indicates the sequence number that is needed for the reader to generate a new shared key to complete the reader-device authentication.

In this design, the reader and the cloud mutually authenticate each other based on the asymmetric key based security scheme. By checking the work schedule, the cloud guarantees that a legitimate worker operating a legitimate reader is assigned the task of reading the data from a specific smart grid device at the right time. Moreover, by matching the smart grid device ID and the location of the reader, the cloud can confirm if the reader is physically close to the smart grid device. Only after the reader is physically close to the smart device, can the reader get the sequence number for the reader-device authenticated, so it minimizes the size of attack window. After getting the sequence number, the smart grid device calls the key generation algorithm (see Section III-D), as denoted as $NewSK()$ in Figure 3, to generate a new shared key with the specific smart grid device with the ID as of M_{id} . Next, we show the details of the reader-device authentication protocol.

C. The Reader-Device Authentication Protocol

Unlike the asymmetric key based protocol used in the reader-cloud authentication, symmetric key based protocol is used for reader-device authentication, because smart grid devices with limited computation power and memory space are usually not able to support computation intensive algorithms like asymmetric key based security schemes [17]. When a symmetric key based protocol is adopted, we have to worry about the distribution and compromise of the shared key. For example, if the reader is lost or stolen, the shared key stored in the reader will get leaked. On the other hand, complicated protocols are usually required to distribute the shared key. We address the above problems by proposing a secure and efficient symmetric key based reader-device authentication protocol.

The detailed protocol is depicted in Figure 4. It consists of two steps. First, a shared symmetric key is generated at both the reader and the smart grid device. Second, a symmetric key

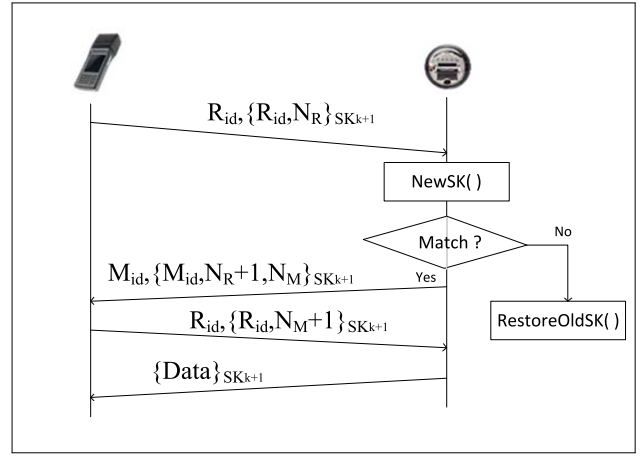


Fig. 4. The reader-device authentication protocol.

based authentication is completed to mutually authenticate the smart grid device with the reader.

We assume that before authentication, a legitimate reader and the smart grid device have a pair of shared key named old shared key, SK_k , which is the key used in the k_{th} reader-device authentication. With the sequence number, the reader can generate a new shared key, SK_{k+1} , by calling the new key generation function $NewSK()$ (described in Section III-D). Then the reader sends the smart grid device a message $\{R_{id}, \{R_{id}, N_R\}_{SK_{k+1}}\}$, where N_R is a random number. After receiving the message, the smart grid device generates a new shared key, SK'_{k+1} , using the same function $NewSK()$. If key SK'_{k+1} cannot successfully decrypt the received message, it means that $SK_{k+1} \neq SK'_{k+1}$ and authentication fails. Thus, the request is ignored and the old key SK_k is restored at the smart grid device. Otherwise, the device continues the authentication process by responding a message, $\{M_{id}, \{M_{id}, N_R + 1, N_M\}_{SK_{k+1}}\}$, where $N_R + 1$ is the response to the challenge from the reader and N_M is a random number to challenge the reader. After the reader receives the message, it decrypts it using the shared key, SK_{k+1} to get the response and challenge from the smart grid device. Then, the reader responds with a message, $\{R_{id}, \{R_{id}, N_M + 1\}_{SK_{k+1}}\}$, to the device. It completes the authentication so that the smart grid device can trust the reader, because it has the shared key. Later on, all the data communication between the reader and the smart grid device is then encrypted using the shared key. To enhance the security in the authentication, we limit the maximum number of requests from the same reader. The reader is blocked after it sends more than Max_Try number of requests.

The key of this authentication protocol is that both the reader and the smart grid device should have the same set of SK_k and SK_{k+1} . Otherwise, the authentication fails. Therefore, how to generate a pair of shared key between the smart grid device and the reader is critical for the success of the authentication. Next, we illustrate how the new shared key is generated.

D. One-Time Symmetric Key Generation

$NewSK()$ is the function to generate a one-time symmetric key shared by the smart grid device and the reader. A hash

function is used as the new key generation function as defined in Equation 1.

$$SK_{k+1} = \text{Hash}(\text{Zipcode}, TMSP_{M_{id}}, Seq\#, SK_k) \quad (1)$$

In the equation, we can see that the input of the key generation function includes the area code of the smart grid device, the timestamp $TMSP$ of the first response message from the device before authentication, a shared sequence number and the previous shared key. Please note that $TMSP$ has granularity of one second while reader-device authentication execution time is less than one second. Because the same hash function is used at both sides of smart grid device and the reader, if the meter and the reader can have the same input, the same new key, SK_{k+1} is then generated at both sides. Initially, when the smart grid device is programmed by OG&E, the initial shared key SK_0 and the initial sequence number is installed in the smart meter. Later on, every time when the data is read from the meter, $SK_1, SK_2, \dots$ are generated accordingly using the key generation function. As a result a key chain is generated as shown in Equation 2 and each key is used only once.

$$Key_0, Key_1, Key_2, Key_3, \dots, Key_k, Key_{k+1}, \dots \quad (2)$$

In Equation 1, we also need to decide the value of the sequence number. The sequence number is initialized as a random integer ranging between 0 and $MaxSeq$, but it increases by one each time when data is read. Even though increasing the $seq\#$ by a random number instead of one can make it harder for attackers to guess it, it also results in message overload to agree on the random number between the cloud and the smart grid device. Additionally increasing the $Seq\#$ by one will not reduce the randomness of the generated shared key a lot knowing that there are three other parameters of the hash function including the random fine granularity timestamp, $TMSP$. The sequence number is updated as in Equation 3, where $MaxSeq$ is the maximum value of the sequence number.

$$Seq\# = (Seq\# + 1) \bmod MaxSeq. \quad (3)$$

E. Discussion

We summarize several security and reliability related features of the proposed framework. Two different types of authentication protocols are used considering the physical limitations of the hardware in the system. Symmetric key based reader-device authentication is much more lightweight than the asymmetric key based authentication. It is supported by ANSI C12 standard [1], which requires that all the smart grid devices should be equipped with the capability of supporting symmetric key and basic hash functions. The design of our framework matches the design principles of resource constrained systems, like smart grids, very well [17]. In our design, the asymmetric key is used for reader-cloud authentication, because readers are much more powerful than smart meters, while symmetric key based authentication is used between the meter and the reader by satisfying the constraints of the meter. In addition, the information used to generate the shared key is distributively stored in a cloud based sequence number service and a legitimate smart grid device, and physical information such as location, time, random initial sequence number and a random timestamp

are used in key generation to further enhance the security of the protocol. Although the old key is also stored in a cloud based key storage service, it is only used as a backup. All these factors significantly increase the efforts needed to guess the generated shared key between the reader and the device. The design of the one-time key approach also addresses the big problem of symmetric key distribution. Symmetric key revoking can be achieved by installing a new key to the smart grid device with a wired communication with the smart grid device when it is necessary. It is easy to be accomplished by considering the nature of the application. Finally, we try to keep the attack window as small as possible, because the attack can start only after the reader is physically close to the smart grid device, and we have to add a time limitation for the length of authenticated session.

In the secure and efficient framework for isolated smart grid device reading, each time a smart grid device needs to be read, a process of two authentication protocols is executed. Since the utility cloud stores all reader and job information, any legitimate reader can get the corresponding $seq\#$ for each smart grid device and get authenticated with the smart grid device after it gets authenticated with the utility cloud. Thus, the many-to-many relationship between the reader and the smart grid device can be correctly supported. Moreover, with the symmetric key generated each time when reading the smart meter, we are actually addressing two issues, avoiding using computation heavy security algorithms, like asymmetric key schemes, and avoiding the difficulty of symmetric key distribution.

Working with the two-way communication system which provides reliable and high communication bandwidth as depicted in Section II, existing security protocols like [20] help secure the cloud system, and smart grid device owners take care of the physical security of the smart grid device. In addition to that, the secure protocol that reads connected smart grid devices and this proposed framework that reads isolated smart devices can significantly improve the security and reliability of the data collection process in smart grid system. Therefore, we can build a secure and reliable smart grid. More discussion on the security and reliability of our design against typical attacks comes in the next section.

IV. SECURITY ANALYSIS

The secure and efficient framework to read isolated smart grid device is proposed to prevent unauthorized parties to read the data collected and stored in some not networked smart grid devices like smart meters. In this section, we formally analyze the security properties of the proposed framework in the context of a set of common attacks to the smart grid system as seen in previous studies [22], [24].

A. Eavesdropping

Eavesdropping is an attack to capture the unauthorized information that is confidential. Encryption provides strong defense against eavesdropping. In our framework, all communications are encrypted to prevent eavesdropping. Theoretically, if the attackers cannot get the key, they cannot decrypt the message to get what is inside the message.

Brute-force attack is a way to get the key, which is analyzed in the next section.

B. Brute-Force Attack

Brute-force attack is to try every possible key on a piece of cipher text until an intelligible translation into plain text is obtained. Since we are using modern cryptographic algorithms like AES, the basic brute-force attacks have been proved to be blocked by using these algorithms with appropriately sized keys like 256-bit keys [34].

The other type of brute-force attack is more sophisticated and tries to guess a new key by knowing the key generation algorithm. In our framework we block this type of brute-force attack by using a total of four parameters in the new key generation function, $NewSK()$ as defined in Section III-D, where in addition to the Zip code, the previous key and the seq#, we use a timestamp with a granularity of one second to generate the new key. The attacker has to try $O(2^{256})$ times to successfully guess the new key as the length of the key is 256, therefore, it is almost impossible for the attacker to deliver a brute-force attack. In case that in addition to the SK_k and the $zipcode$, the timestamp is exposed to a hacker, the probability of guessing the key depends on the randomness of the last parameter, $seq\#$. The probability is $\frac{1}{MaxSeq}$, so if we use a big value of $MaxSeq$ in our framework, the probability can be very low. In addition, distributed storage of parameters has a positive effect to reduce the effectiveness of attacks, since some parameters like $seq\#$ are from the cloud, while others, like the timestamp, are from the smart device. Moreover, we limit the number of allowed authentication requests to relieve us further from brute-force attacks. As a result, we can say with a high level of confidence that brute-force attack can be mostly blocked in our framework. Finally, because the shared one-time keys (SK_k) in the key chain have no predictable relations among them, it is not helpful for the attackers to guess the next round key even though they can have access to multiple previous keys.

C. Man-in-the-Middle Attack

A Man-In-The-Middle (MITM) attack is the attack in which the attacker pretends to be the right person during the communication and uses the information received from one side to fool the other side. Both sides of victims feel that they are exchanging information directly. In the reader-cloud authentication, public/private key pair based digital signature is used in mutual authentication. It guarantees a strong defense against MITM attacks, because an MITM attacker is not possible to forge the digital signature without the right private key, except that the attacker convinces the cloud accepting his public key. However, in smart grid architecture, the reader is a part of the cloud before it goes out for work, so an invalid public key cannot be introduced to the cloud. In the reader-device authentication, a symmetric key is shared between the reader and the smart device. Anyone who does not have the key cannot win the trust from either the reader or the smart grid device. One type of malicious reader may launch the following attack. In the first message of the reader-cloud authentication,

the reader sends $\{R_{id}, T_R\}_{key}$, $\{key\}_{Pub(c)}$, and $\{R_{id}, T_R\}_{Pri(R)}$. If the attacker pretends to be a legitimate reader and sends the first message by replacing the one time session key to be key' , the attacker can generate a message $\{R_{id}', T_{R'}\}_{key'}$ and $\{key'\}_{Pub(c)}$, but without generating a valid signature, the cloud detects the loss of data integrity by comparing the digital signature; therefore it drops the connection. In the worst case, even if the attack can get a valid ID, it still needs to go through the job verification in the second step of the authentication protocol, i.e., if the attacker who has a valid ID but not assigned to complete the job, he cannot get the trust from the cloud so that he cannot get the $Seq\#$ to generate the new shared key in the reader-device authentication. Above discussion has shown that the brute-force attack is mostly impossible to get the shared key. Finally, short range secure communication, like ZigBee, makes the MITM attack even more difficult. Therefore, we conclude that the MITM attack can be blocked.

D. Device Attack

A device attack aims to compromise or mimic a legitimate device, such as a smart meter and a reader. In our design, a fake reader has no way to get authenticated, because it cannot get a valid private key and attacks from a compromised reader can be treated as the same as a type of internal attack which is discussed in next section. A fake smart grid device cannot get authenticated with the reader and tamper-proof techniques can be used to prevent modifications to the smart device.

E. Internal Attack

An internal attack happens when an attacker is from the organization or has the assets that can help him to access the unauthorized resource. Our analysis for internal attacks consists of two scenarios, a lost legitimate reader and malicious attacks from inside people who are legitimate workers. The authentication request from a lost legitimate reader can be blocked, because the reader cannot provide the right task information, so a lost reader can only read the smart device for the rest of the authenticated session, which is a very small window like 30 minutes. Further readings can be blocked, because it cannot get the new shared key. Limited information accessing by the lost reader is negligible for privacy concerns. In the second scenario, the cloud verifies the reader based on the provided task information, e.g., the matching of location and verification on the working schedule, so as to disqualify the attacker who has only login information, but is not assigned to process this specific task.

F. Replay Attack

In a replay attack, an attacker gains the information between two victims and intercepts the information and replays it fraudulently. This attack can be mostly disabled by the use of one-time shared key in the reader-device authentication. In addition, both authentication proposals use timestamps and nonce to prevent replay attack. If the attack wants to achieve his goal, he has to guess the one-time shared key and nonce, but as analyzed above, it is almost impossible.

G. Forward Secrecy

Forward secrecy ensures that even if one session key gets compromised, other session keys can not be compromised. Our shared key is a session key. Every new task requires both the reader and the smart grid device to generate a new shared key for authentication. Therefore, the compromise of one shared key will not cause the compromise of others without knowing three other parameters used in new key generation. The shared keys except the latest one can be removed periodically from the reader and stored in a cloud based old key storage service for backup. The design of the framework makes it almost impossible to guess the new shared key, because the new key is generated based on the randomness in the location and the timestamp, i.e., having the last old key, the probability successfully guessing the new key is only $\frac{1}{MaxSeq \times 2592000}$, where $MaxSeq$ is the maximum sequence number and 2592000 is the possible number of timestamps, even assuming that the attacker knows the Zip code and the reading pattern of the worker to read the meter like one reading per month. In addition, knowing more than one previous key will not increase the probability of successful guess, because each key is generated independently. Thus, this framework has perfect forward secrecy.

H. Denial-of-Service Attack

In a Denial-of-Service (DoS) attack, an attacker attempts to prevent legitimate users from accessing information or services, by targeting the hosting device. In our framework if a hacker tries to attack reader-device authentication, there will not be much harmful effect, because all rounds of message exchange between the reader and the device are encrypted using a key shared between them, and any message without this encryption can be neglected. The other target for DoS attack could be the cloud, but the cloud located in utility company's data center is usually well protected by using the latest security technology, and the cloud can ignore the first message from the reader if it is not signed using the eligible reader's private key, or if it has a wrong user ID. Therefore our framework is mostly secure against Denial-Of-Service attack.

V. IMPLEMENTATION AND EVALUATION

Having analyzed the security properties of our framework, we evaluate its performance by a simple prototype implementation. We mainly evaluate the cost of different hash algorithms and cryptography algorithms, as well as the cost of the proposed framework. Smart devices such as iPhones, iPods, and iPads are used to simulate the reader and the smart meters. An iMac desktop computer is also used in order to show the impact of the device capability on the performance of security algorithms. The running time of the algorithms is used as the metric to evaluate the cost.

A. Experiment Setup

We implement the prototype of the proposed secure framework by building an integrated system with three parties, the cloud, the reader and the device. Two iOS applications are

	OS	Processor	Memory
iPod Touch	iOS 8.1.1	Dual core, 1000 MHz	512 MB
iPad Mini	iOS 7.1	Dual core, 1000 MHz	512 MB
iPhone 4S	iOS 7.1.2	Dual core, 800 MHz	512 MB
iMac	OS X Yosemite	Intel Core 2 Duo 2.4 GHz	4 GB

Fig. 5. Reader specifications.

developed to simulate the smart reader and the isolated smart grid device respectively. A Java application is implemented to simulate the cloud. It has a connection with a database that stores job information and implements a RESTful Web service to handle the authentication request from the reader. The cloud is built based on several java libraries including java.security, javax.crypto, and bouncycastle.crypto. All these applications work together as a prototype of the proposed framework. We use three different smart readers, including an iPad mini (first generation), an iPod touch (5th generation) and an iPhone 4S, as well as an iMac desktop computer for testing and comparing the results. The main specifications including the version of OS, the speed of processor and the size of memory for these devices are denoted in Figure 5. Xcode 6 with libSystem CommonCrypto library support is used to build the iOS applications, and Netbeans 8.0.2 with JavaSecurity package in Java SE 7 is used to build the cloud service application. The iMac runs OS X v10.5 and the iOS devices run iOS 5.0 or later. The iMac is more powerful than the three iOS devices and the three iOS devices have similar configuration, but iPhone 4S has a relatively weaker configuration among these three.

B. Efficiency Comparison of Hash Algorithms

In the reader-device authentication, hash function is used to generate the one-time symmetric key. Thus, we evaluate the cost of different hash functions in the context of running time using different devices. Four of the most popular hash algorithms are evaluated, including MD4, MD5, SHA1 and SHA2 as shown in Figure 6 and Figure 7. Figure 6 (a), (b), and (c) show the cost of the hash function in iPod Touch, iPad Mini, and iPhone 4S respectively, while Figure 7 shows the cost of the hash function in iOS simulator running on an iMac desktop computer.

In the experiment, hash functions are used to generate the hash code of a variant size of data of 10KB, 100KB and 1MB. The running time to get the hash code is recorded as the average of 100 times of repetitions. The X-axis of each figure shows the size of the data and the Y-axis depicts the running time. By comparing the execution time of the hash algorithms running in the iOS Simulator and the actual smart devices, we observe the impact of the device capability in executing hash algorithms, i.e., we notice a remarkable saving in execution time when using a powerful device, like a iMac desktop computer, which uses about one-fifth of the time to generate the hash code of the same size data compared with the smart devices. We also observe that the cost of hashing increases almost linearly with the size of the data. All four hash algorithms are efficient and have close performance in terms of the running time using smart devices, which are about 2ms

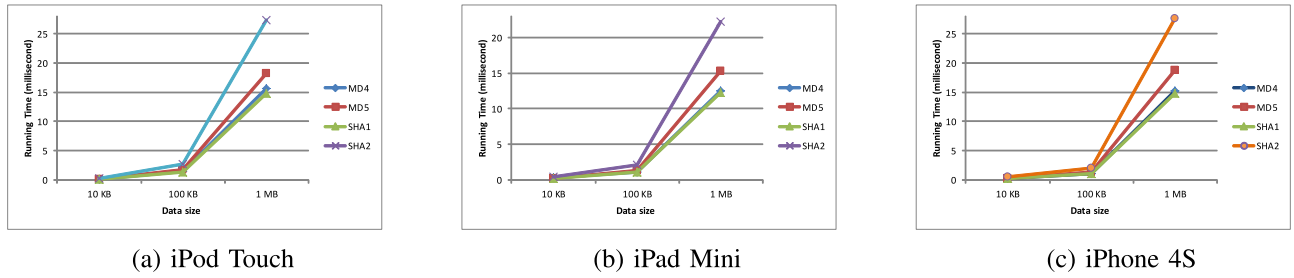


Fig. 6. Execution time of hash algorithms using iOS smart devices.

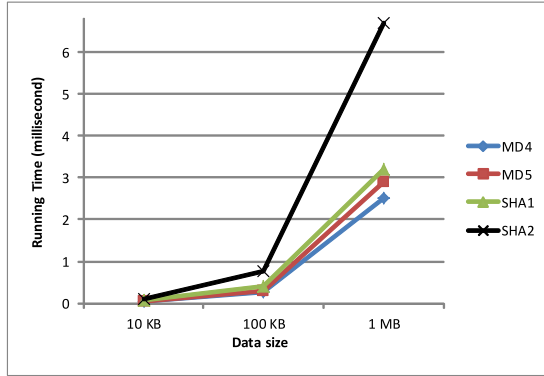


Fig. 7. Execution time of hash algorithms using iOS simulator in iMac.

for getting hash code of 100KB data and about 23ms for 1MB data. Thus, we can believe that even in a less powerful smart grid device than iPod and iPhone, the hash based symmetric key regeneration will not cost a lot and could have an acceptable performance. Among all four hash algorithms, SHA1 has a relatively low cost especially when the size of the data is getting larger, and it is at least as secure as other hash algorithms, so SHA1 is recommended to be used in our framework implementation.

C. Efficiency Comparison of Cryptography Algorithms

Symmetric cryptographic algorithms are used in the reader-device authentication, while asymmetric cryptographic algorithms are used in the reader-cloud authentication. Here we evaluate the performance of three widely used symmetric cryptographic algorithms (AES, DES and 3DES) and one asymmetric cryptographic algorithms (RSA with key size of 2048 bits) using iOS stimulator running on iMac desktop computer, iPad Mini, iPod Touch, and iPhone 4S. Each algorithm is tested to encrypt and decrypt different sizes of data of 10KB, 100KB and 1MB. The time that it takes to encrypt and decrypt different sizes of data is recorded as the average of 100 times of repetitions. Figure 8 (a) and (b) display the execution time using iOS simulator in iMac for encryption and decryption operations respectively. Figure 9 (a), (b) and (c) show the cost of decryption operation for iPod Touch, iPad Mini, and iPhone 4S respectively, and Figure 10 (a), (b) and (c) depict the cost of decryption operation for iPod Touch, iPad Mini, and iPhone 4S respectively. The X-axis of each figure shows the name of the algorithm and the Y-axis depicts the running time of algorithms.

From all figures, we observe that the running time of the encryption and decryption increases almost linearly with the increase of data size in all algorithms, and RSA algorithm, in general, takes significantly more time in both encrypting and decrypting the same amount of data than the other three symmetric cryptographic algorithms, i.e., RSA algorithm is about thousands times slower than others. This difference is even more significant in a less powerful device, like a smart meter, as we can observe that there is a noticeable difference in the execution time of the cryptographic algorithms running in iOS simulator in iMac compared with the smart devices due to the capability differences. Among the three symmetric cryptographic algorithms, AES, DES and 3DES, AES seems to have a slight better performance and a better security, so it is recommended to be used in our framework. The evaluation results validate the argument that symmetric cryptographic algorithms are much more preferred in resource constrained systems than asymmetric cryptographic algorithms.

D. Performance of the Framework

We implement a prototype of the proposed framework by choosing the following security algorithms, RSA with key length of 2048 bits as the asymmetric cryptographic algorithm, AES with key length of 128 bits as the symmetric cryptographic algorithm, and SHA1 as the hash function in the new key generation. The implementation consists of three applications as described in the experiment setup.

In the OG&E smart grid system, every 15 second the smart meter collects about 24 bits data, which includes the electricity consumption value, the timestamp, and the meter ID, so we prepare a set of 4 MB randomly generated data as an estimation of the data collected from a single smart meter every month. The collected data will be temporary stored in the flash memory of smart meter before it is transmitted. For example, OG&E can use GE I-210+c [13] Smart Meter, which has a 32-bit ARM processor, 8 MB RAM, 4 MB flash memory and supports ZigBee protocol, in their smart grid implementation.

The main functions of the cloud application are to verify the reader, check the task information, and provide the encrypted sequence number for the reader to generate the shared key. This application is running on a PC connected to the same wireless network as the other applications for the reader and the smart device. The reader application has two major functions, connecting to the cloud and getting the sequence number, and connecting to the smart grid device

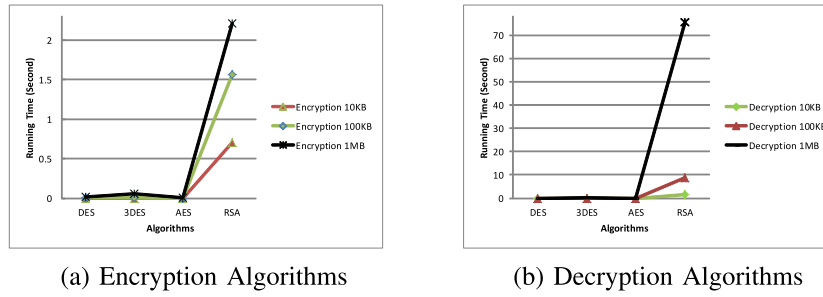


Fig. 8. Execution time of encryption & decryption algorithms using iOS simulator in iMac.

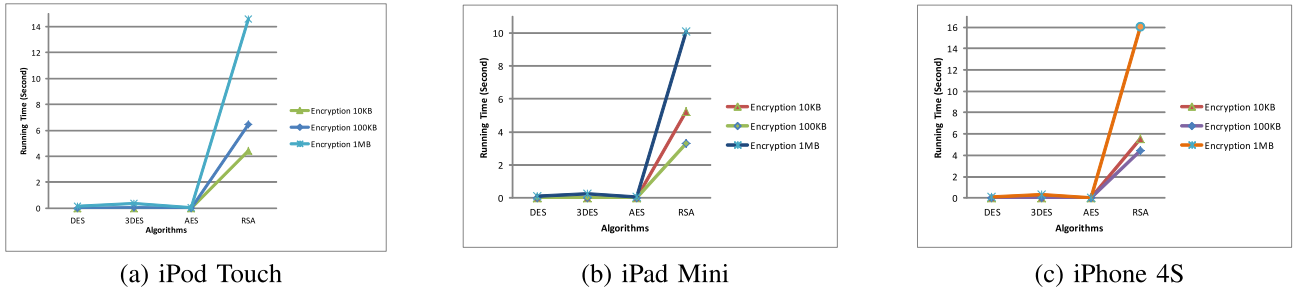


Fig. 9. Execution time of encryption algorithms using iOS smart devices.

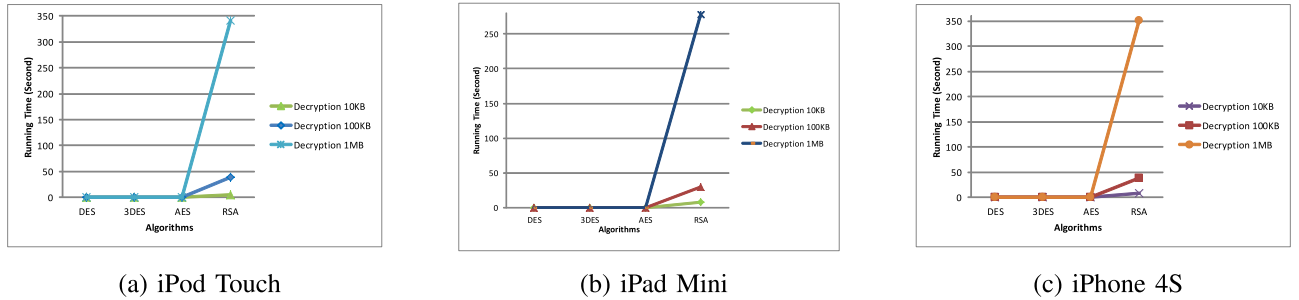


Fig. 10. Execution time of decryption algorithms using iOS smart devices.

application and reading data. The smart grid device application is an iOS application simulating a smart grid device.

We divide the running time of the framework into three parts and the running time of each part is measured and recorded. The first part includes the first two steps in the framework, and the second part covers the reader-cloud authentication while the last part specifies the reader-device authentication and data reading. We fragment the 4 MB data before it is transmitted, considering the packet format of ZigBee protocol, which has a packet size of 127 bits, and transfer these packets one by one.

The execution time of three different parts of the framework as well as the total amount of time is shown in Figure 11. The first three groups of bars specify the execution time of three parts of framework and the fourth group of bars denotes the total amount of framework execution time. Each group of bars shows the execution time in different devices in the order of iPhone 4S, iPod Touch, iPad Mini and iOS simulator in iMac. In the figure, we find that iOS simulator has a much shorter execution time than the smart devices because of the capability difference. We also observe that the second part takes almost half of the total execution time of the framework prototype, which justifies the argument that

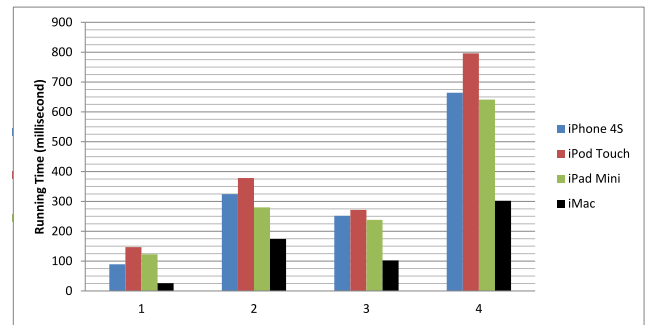


Fig. 11. Framework implementation for 205 KB data.

asymmetric cryptographic algorithms are much more costly than symmetric cryptographic algorithms, even when used only in simple authentication protocols that do not have a large quantity of data to be encrypted and decrypted. As for three iOS smart devices, we notice that the iPad Mini has a shorter execution time because of relatively powerful configurations. This experiment shows that the whole process of the proposed framework can be completed within one second

using the smart devices and it validates the applicability of the framework in smart grid.

VI. RELATED WORK AND DISCUSSIONS

Having introduced our framework for securely reading data from smart grid devices, in this section, we list a set of previous efforts that share similar goals as ours.

Advanced metering infrastructure (AMI) that automatically measures and collects power grid system data has been of great interest. Several projects have been launched such as [6], [7], [9], [15], [16], [25], [26], and [38] targeting to build AMI. SCADA [4] has been built to automatically read utility data. Various types of smart meters are also designed for a similar purpose. Except for smart meters, the other very important component of AMI is the communication network. There are many ways to build the two-way communication system. Technologies such as power line carrier (PLC) [12], broadband over power lines (BPL) [3], and others [9], [38] have been extensively explored. The communication system described in this framework uses a different technology to achieve the goal of higher bandwidth and better security.

One of the major goals of smart grids is to improve the reliability of the traditional power grid by integrating high-speed and two-way communication technologies [37]. Security plays a vital role in building a high reliable communication system for smart grid. Several papers [11], [17], [23], [37] all argue that because of the hardware constraints and high real-time requirements of smart grid applications, lightweight security protocols are extremely important and welcomed. Technologies such as symmetric key and one-time key are preferred. Our work matches all those design principles very well. It is also agreed that confidentiality of the data in smart grid is critical to the smart grid system. A lot of research [18], [27], [28], [32], [36] has shown the high risk consequence of the leakage of the energy usage data.

Authentication is one of the most important techniques to prevent sensitive data leakage. Most existing work on authentication protocols in smart grid focuses on authentication problems in smart grid communication networks. A lightweight message authentication scheme is proposed in [11], where a Diffie-Hellman exchange protocol is used to establish mutual authentication among meters and establish a shared session key. Later on, the smart meters use the shared session key to authenticate subsequent messages. Ayday and Rajagopal [2] propose a set of authentication mechanisms between the devices in the home area network, but they assume that a pair-wise symmetric key is existing between devices and the center of trust. An identity-based signcryption is proposed in [33] to provide a zero-configuration encryption and authentication for smart grid. The paper illustrates an interesting idea of asymmetry key management mechanism by replacing certificate authority (CA) with a key-generating server (KGS). Instead of making the message sender to talk to CA to distribute the keys, the paper asks the recipient of the message to obtain necessary key from the KGS. All above efforts try to establish the authentication between the different components in the smart grid communication network.

The secure and efficient framework proposed in the paper works in a different scenario from above protocols and it provides a more lightweight solution than above solutions by only using symmetric key in the reader-device authentication.

DLMS/COSEM is another popular approach used to model smart meter functionality and it can add new data security elements [19]. In the proposed DLMS/COSEM security scheme, the low level security model is a password based authentication so it is not secure enough, while the high level security model suffers from a key distribution problem, which can expose a session secret key that is supposed to be shared only by two devices to a third party node [5]. In addition, DLMS/COSEM security scheme is more suitable to be used in a networked environment. The proposed protocol in this paper generates one-time key, which is more secure and is designed for securely reading isolated smart grid devices.

Kerberos [30] is a widely used secure system that makes use of symmetric-key based authentication. It has been proven to be efficient and scalable. A ticket-granting service is used to support subsequent authentication. Kerberos-based authentication may be used for the authentication between the reader and the utility cloud, but it may not be suitable to be used in the reader-device authentication; otherwise, it has to require the smart grid device to verify the ticket from the reader, but with very limited resource and knowledge at an isolated smart grid device, it will be a big challenge. Kerberos also has to assume a shared key between the ticket grant server and the smart grid device. If the shared key is breached, the protocol is no more secure. In addition, Kerberos has strict time requirements, which means that the clocks of the involved hosts must be synchronized within configured limits. Finally, the logic in Kerberos is more complicated than the one-time key based approach proposed in this paper.

One-time key based approach has also been studied in some previous work. Nabeel *et al.* design a key management scheme [29] to secure end-to-end communication in the advanced metering infrastructures (AMI). The strong authentication in their system is built based on the use of the physically unclonable function (PUF) devices, which can generate and regenerate symmetric keys and access level passwords for smart meters. Similarly, SCAPACH [35] is proposed to securely collect telemetric data from a large scale of pole devices in grid systems. It also makes use of a PUF component in their system. Compared with above two solutions, our framework does not rely on a PUF device to re-generate the symmetric key chain, i.e., it is not depending on any special hardware. Both SCAPATH and our proposal work on securely reading sensitive data from isolated devices in smart grid, but we adopt different approaches to establish authentication. One-time signature has been proposed to provide multicast authentication in [21]. Li and Gao [21] improve previous one-time signature scheme by reducing the signature size and the storage cost.

VII. CONCLUSION

With more fine granularity energy usage data collected in smart grids, how to prevent the sensitive data from unauthorized reading becomes an imperative issue in the

design of smart grid systems. This paper proposes a secure and efficient framework to read sensitive data from smart grid devices that are not directly connected with the smart grid data communication network. Based on the security analysis, the proposed framework has been shown to be lightweight and secure, which achieves the design principles of smart grid system protocol design.

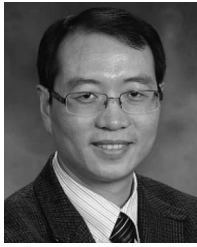
In the current design, the reader has to go through the authentication process with each individual smart device one by one. It may not be very efficient with a big number of smart devices. Thus, an extended framework that can securely authenticate the reader to an area of many smart devices will be explored. Moreover, we will work with OG&E to extend the implementation of the prototype framework, and test and evaluate the framework in a real smart grid environment. Finally, the proposed framework itself cannot detect faulty readings from the smart devices. An extra algorithm is needed to detect faulty smart device readings.

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